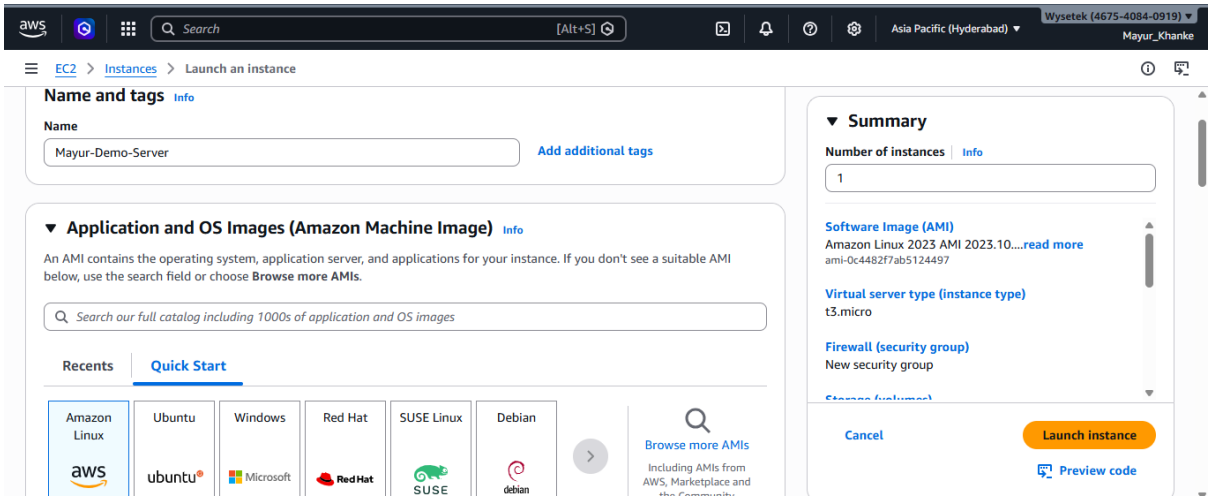
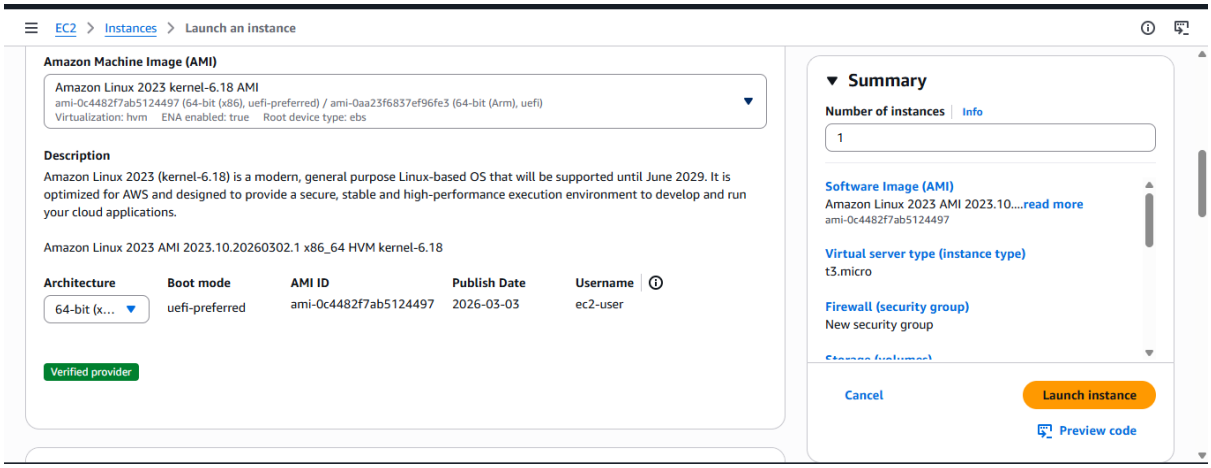
Production Build Challenge

Application Layer Deployment: - This deployment uses AWS services including VPC, EC2, Internet Gateway, NAT Gateway, subnets, route tables, security groups, IAM, AWS Certificate Manager, Application Load Balancer, target groups, Auto Scaling group, and launch templates to build a secure, scalable, and highly available architecture.

I have created a VPC, subnets, Internet Gateway, NAT Gateway, route tables, and security groups using Terraform.

Step 1:- Create a ec2 instance deploy it at private subnet.





EC2 > Instances > Launch an instance

Instance type

t3.micro — Free tier eligible
Family: t3 2 vCPU 1 GiB Memory Current generation: true
On-Demand RHEL base pricing: 0.04 USD per Hour
On-Demand Windows base pricing: 0.0204 USD per Hour
On-Demand Ubuntu Pro base pricing: 0.0147 USD per Hour
On-Demand Linux base pricing: 0.0112 USD per Hour
On-Demand SUSE base pricing: 0.0112 USD per Hour

All generations

Compare instance types

Additional costs apply for AMIs with pre-installed software

Key pair (login) Info

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - *required*

mayur-server-demo

Create new key pair

Network settings Info — Edit

Summary

Number of instances | Info: 1

Software Image (AMI)
Amazon Linux 2023 AMI 2023.10....read more
ami-0c4482f7ab5124497

Virtual server type (instance type)
t3.micro

Firewall (security group)
New security group

Cancel — Launch instance — Preview code

EC2 > Instances > Launch an instance

Network settings Info

VPC - *required* | Info

vpc-0c38678a29f0c4a92 (simple-vpc)
10.0.0.0/16

Subnet | Info

subnet-026b6e180d6a68ffc — private-1
VPC: vpc-0c38678a29f0c4a92 Owner: 467540840919
Availability Zone: ap-south-2a (aps2-az1) Zone type: Availability Zone
IP addresses available: 250 CIDR: 10.0.3.0/24)

Create new subnet

Auto-assign public IP | Info

Enable

Additional charges apply when outside of free tier allowance

Firewall (security groups) | Info

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

Create security group — Select existing security group

Common security groups | Info

Summary

Number of instances | Info: 1

Software Image (AMI)
Amazon Linux 2023 AMI 2023.10....read more
ami-0c4482f7ab5124497

Virtual server type (instance type)
t3.micro

Firewall (security group)
terraform-20260323050106419900000001

Cancel — Launch instance — Preview code

EC2 > Instances > Launch an instance

Firewall (security groups) | Info

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

Create security group — Select existing security group

Common security groups | Info

Select security groups

Compare security group rules

terraform-20260323050106419900000001 sg-00326a8a10473141d
VPC: vpc-0c38678a29f0c4a92

Security groups that you add or remove here will be added to or removed from all your network interfaces.

Advanced network configuration

Storage (volumes) Info — Simple

EBS Volumes — Hide details

Volume 1 (AMI Root) (Custom)

Summary

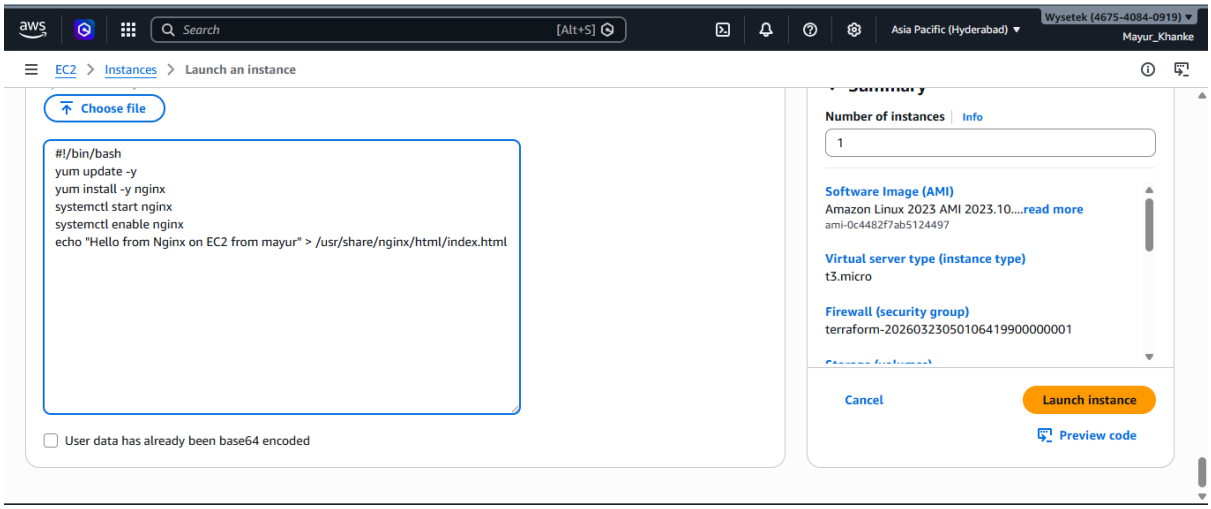
Number of instances | Info: 1

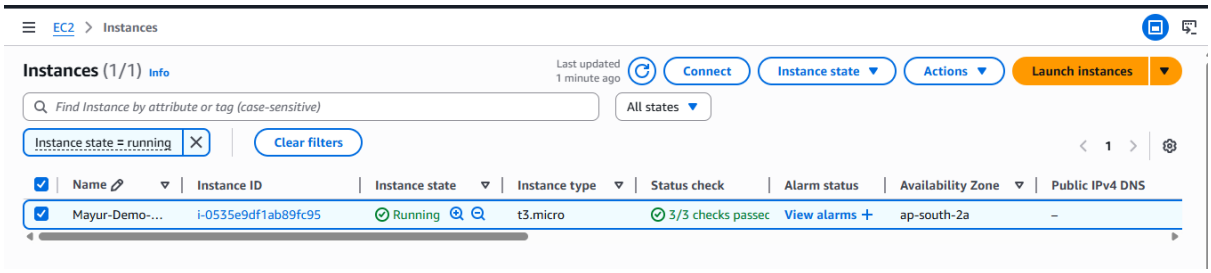
Software Image (AMI)
Amazon Linux 2023 AMI 2023.10....read more
ami-0c4482f7ab5124497

Virtual server type (instance type)
t3.micro

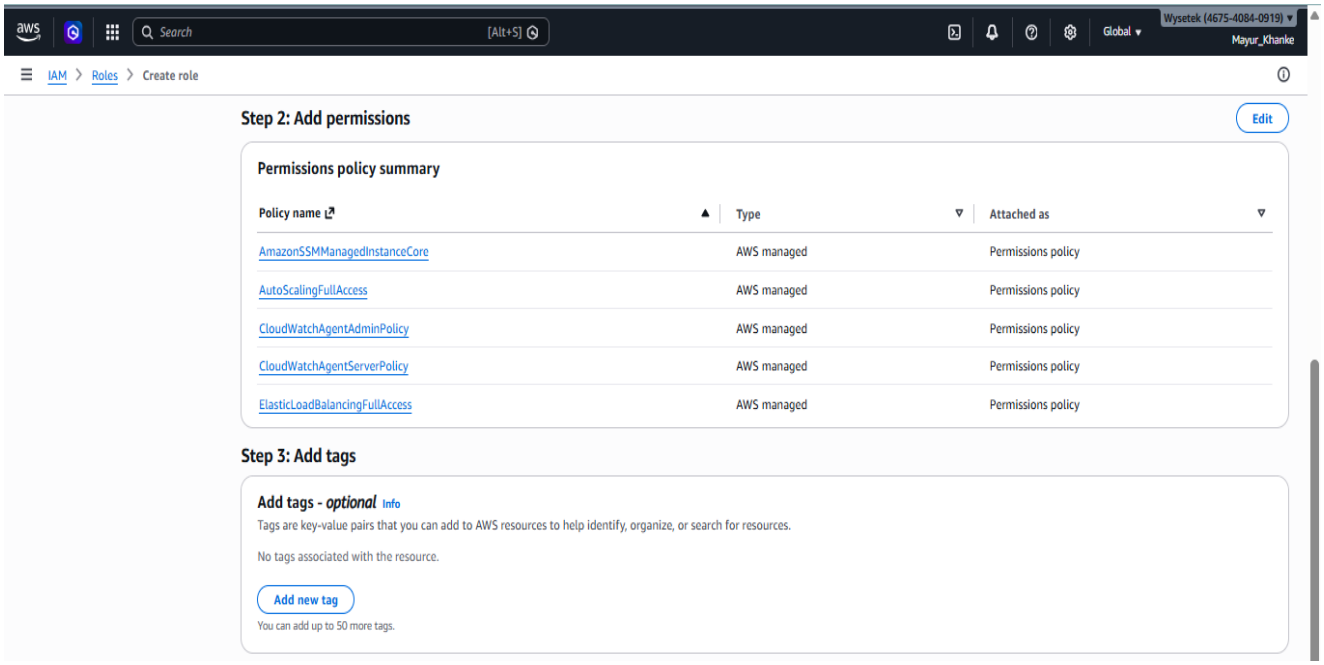
Firewall (security group)
terraform-20260323050106419900000001

Cancel — Launch instance — Preview code

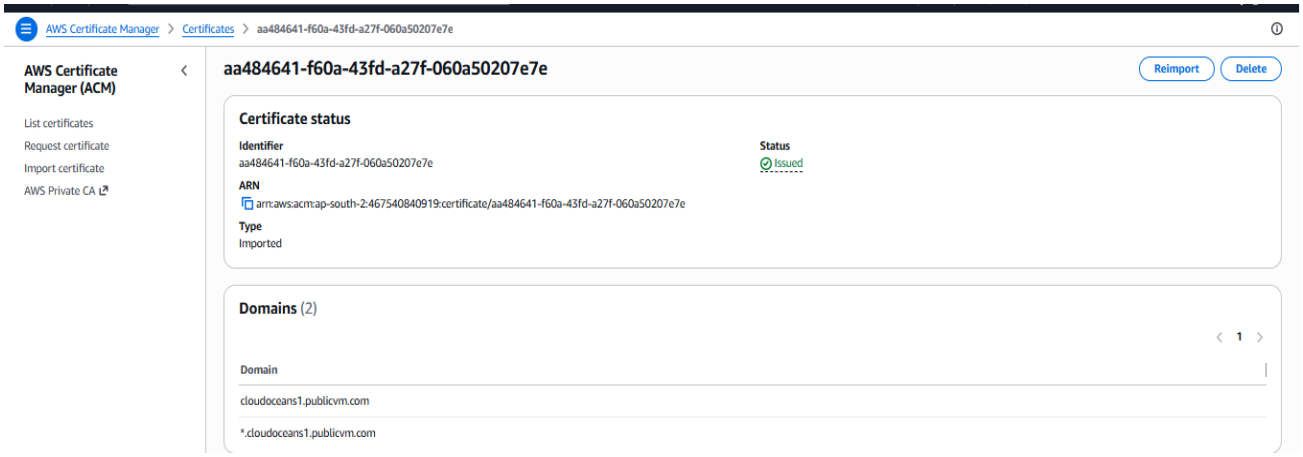




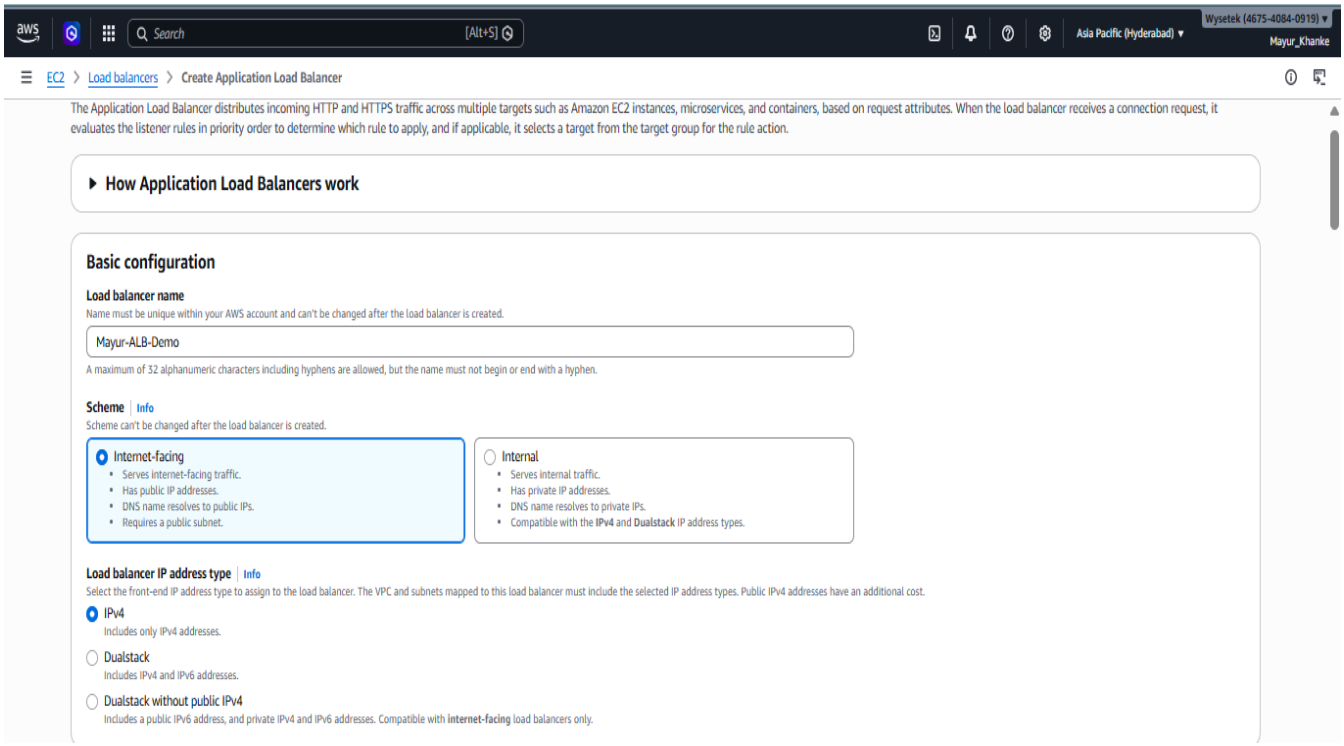
Step 2: - Create an IAM Role and attach this role to ec2 instance.

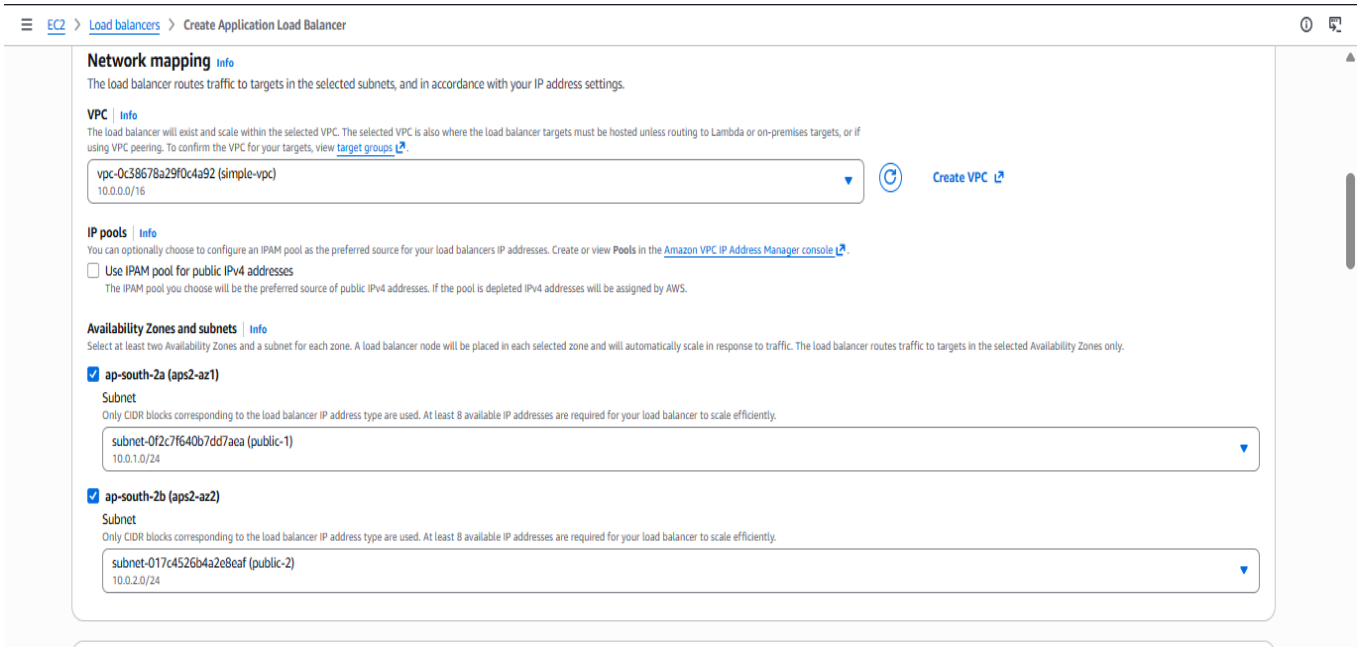


Step 3: - We import the domain certificate in Certificate Manager.

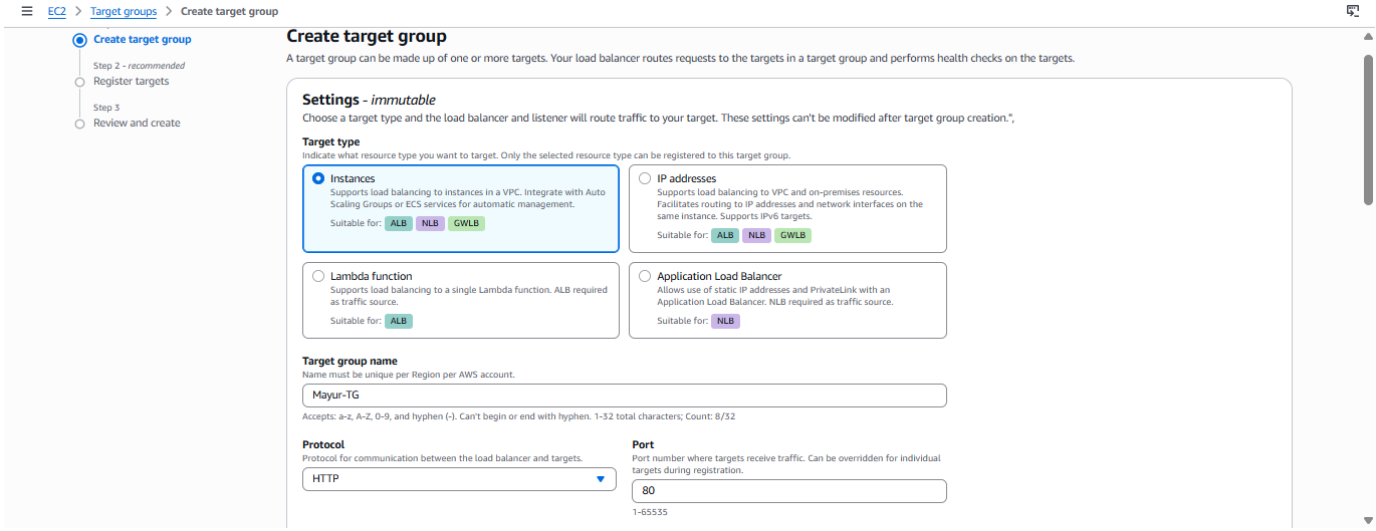


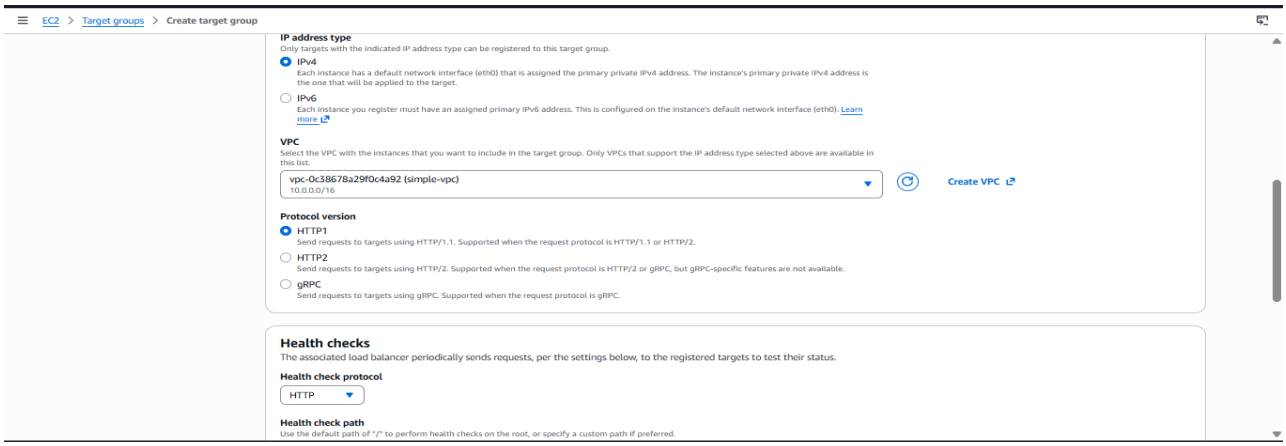
Step 4: - I created an internet-facing Application Load Balancer, configured a target group, set up an HTTPS listener rule, and attached a domain to enable secure traffic routing.



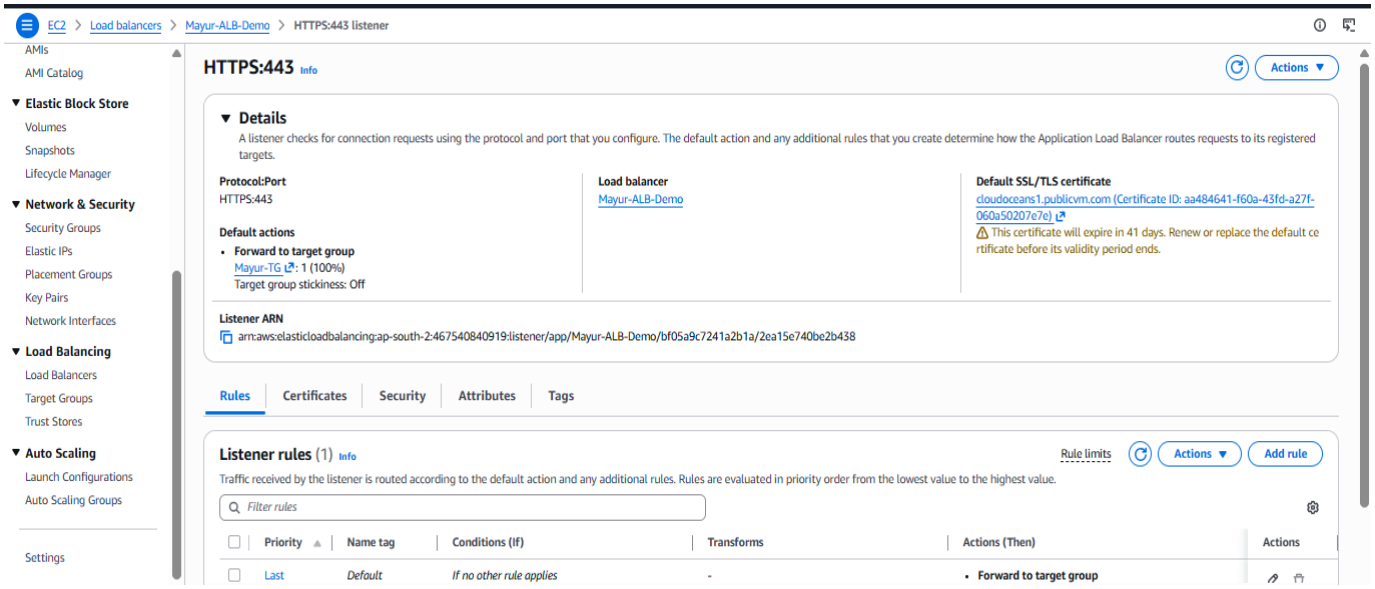


Step 5: - I configured a target group on port 80, attached it to an Auto Scaling group, and enabled it to distribute incoming traffic to dynamically launched EC2 instances.

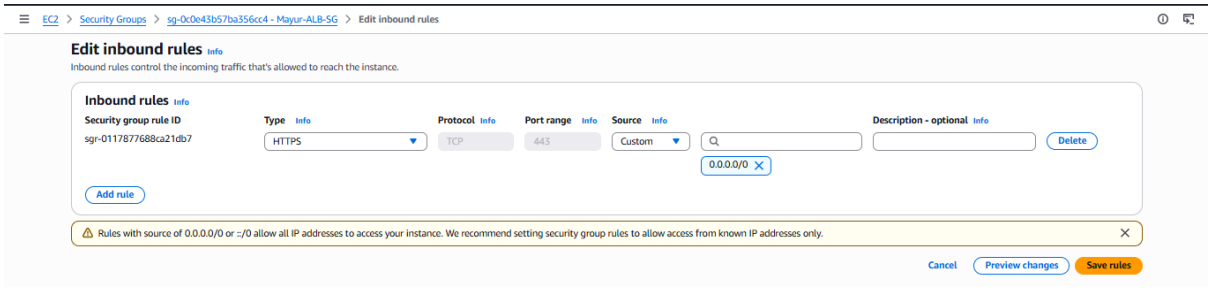




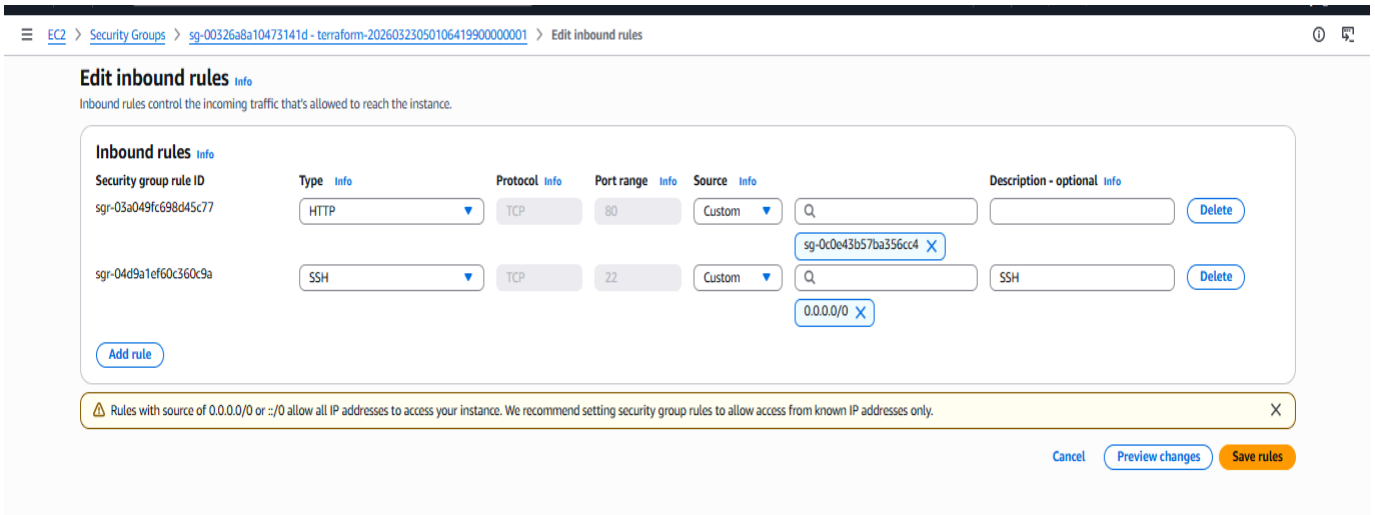
Step 6: - I create a Listener rules for https and attach the target group.

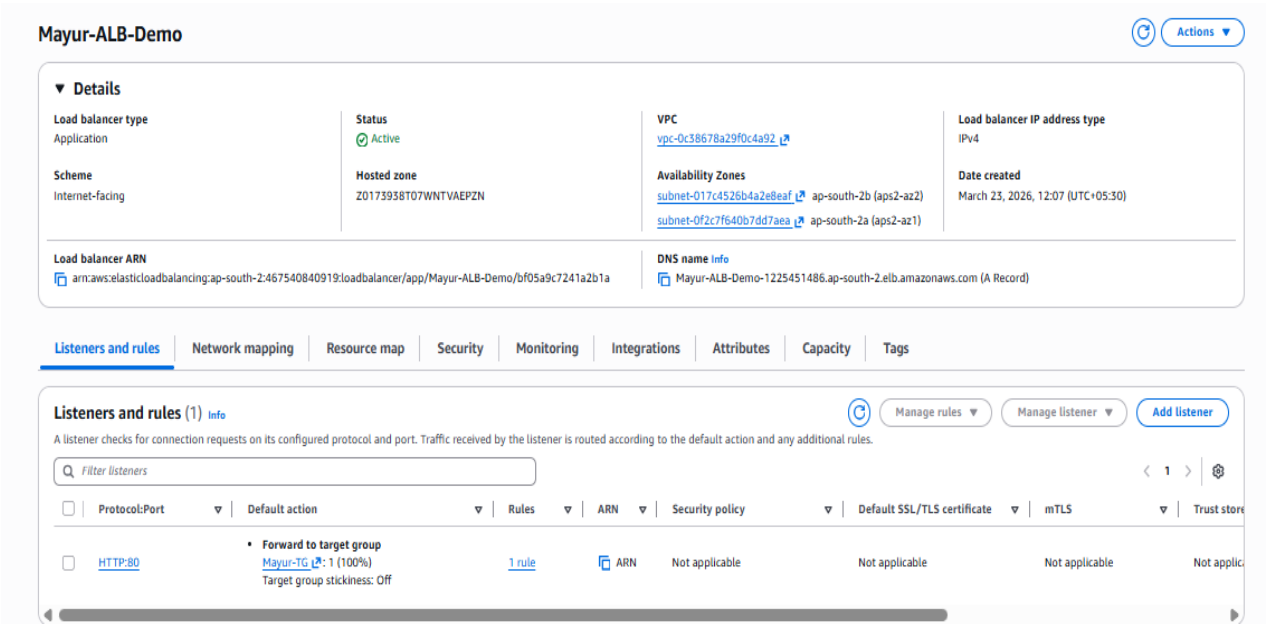


Step 7:- I created a security group for the Application Load Balancer and allowed HTTPS traffic.

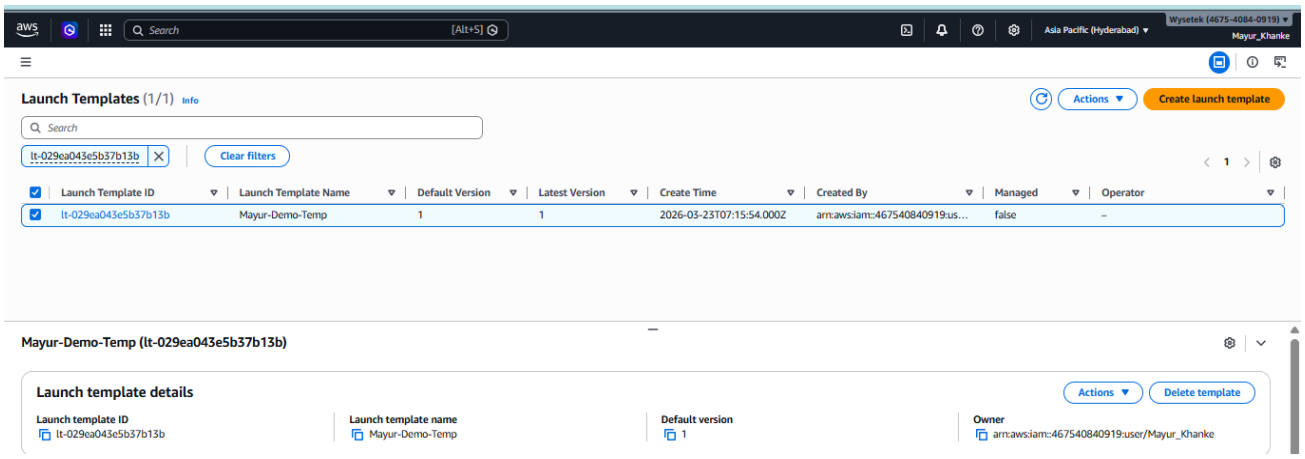


Step 8: - I updated the EC2 security group to permit inbound HTTP (port 80) traffic specifically from the Application Load Balancer security group, ensuring secure and controlled access.





Step 9: - We create a launch template for the Auto Scaling group to define instance configuration.



Step 10: - I created an Auto Scaling group with a desired, minimum, and maximum capacity of 2, attached a launch template, integrated it with an existing Application Load Balancer, and two private subnets.

EC2 > Auto Scaling groups > Create Auto Scaling group

Step 1
Choose launch template or configuration
Step 2
Choose instance launch options
Step 3 - optional
Integrate with other services
Step 4 - optional
Configure group size and scaling
Step 5 - optional
Add notifications
Step 6 - optional
Add tags
Step 7
Review

Choose launch template or configuration Info

Specify a launch template that contains settings common to all EC2 instances that are launched by this Auto Scaling group. If you currently use launch configurations, you might consider migrating to launch templates.

Name

Auto Scaling group name
Enter a name to identify the group.
Mayur-ASG-Demo
Must be unique to this account in the current Region and no more than 255 characters.

Launch template Info

Switch to launch configuration

Launch template
Choose a launch template that contains the instance-level settings, such as the Amazon Machine Image (AMI), instance type, key pair, and security groups.
Demo-Memory
Create a launch template

Version
Latest (1)
Create a launch template version

Description	Launch template	Instance type
-	Demo-Memory lt-0b8eb37340f353af6	t3.small
AMI ID	**Security groups**	**Request Spot Instances**
ami-0fab02efd99dfcbbd	-	No
Key pair name	**Security group IDs**	
FW KP	sg-07d0d593d73328ea1	

EC2 > Auto Scaling groups > Create Auto Scaling group

Choose launch template or configuration
Step 2
Choose instance launch options
Step 3 - optional
Integrate with other services
Step 4 - optional
Configure group size and scaling
Step 5 - optional
Add notifications
Step 6 - optional
Add tags
Step 7
Review

Choose instance launch options Info

Choose the VPC network environment that your instances are launched into, and customize the instance types and purchase options.

Instance type requirements Info

Override launch template

You can keep the same instance attributes or instance type from your launch template, or you can choose to override the launch template by specifying different instance attributes or manually adding instance types.

Launch template	Version	Description
Mayur-Demo-Temp lt-029ea043e5b37b13b	Default	Mayur-Demo-Temp

Instance type
t3.micro

Network Info

For most applications, you can use multiple Availability Zones and let EC2 Auto Scaling balance your instances across the zones. The default VPC and default subnets are suitable for getting started quickly.

VPC
Choose the VPC that defines the virtual network for your Auto Scaling group.
vpc-0c38678a29f0c4a92 (simple-vpc)
10.0.0.0/16
Create a VPC

Availability Zones and subnets
Define which Availability Zones and subnets your Auto Scaling group can use in the chosen VPC.
Select Availability Zones and subnets
aps2-az1 (ap-south-2a) | subnet-026b6e180d6a68ffc (private-1)
10.0.3.0/24
aps2-az2 (ap-south-2b) | subnet-0ec4909808bc71581 (private-2)
10.0.4.0/24

EC2 > Auto Scaling groups > Create Auto Scaling group

Step 1
Choose launch template or configuration

Step 2
Choose instance launch options

Step 3 - *optional*
Integrate with other services

Step 4 - *optional*
Configure group size and scaling

Step 5 - *optional*
Add notifications

Step 6 - *optional*
Add tags

Step 7
Review

Integrate with other services - *optional* Info

Use a load balancer to distribute network traffic across multiple servers. Enable service-to-service communications with VPC Lattice. Shift resources away from impaired Availability Zones with zonal shift. You can also customize health check replacements and monitoring.

Load balancing Info

Use the options below to attach your Auto Scaling group to an existing load balancer, or to a new load balancer that you define.

Select Load balancing options

- No load balancer — Traffic to your Auto Scaling group will not be fronted by a load balancer.
- Attach to an existing load balancer — Choose from your existing load balancers.
- Attach to a new load balancer — Quickly create a basic load balancer to attach to your Auto Scaling group.

Attach to an existing load balancer

Select the load balancers to attach

- Choose from Application or Network Load Balancer target groups
- Choose from Classic Load Balancers

Application or Network Load Balancer target groups

Only instance target groups that belong to the same VPC as your Auto Scaling group are available for selection.

Select target groups

Mayur-TG | HTTP
Application Load Balancer: Mayur-ALB-Demo

Health checks

Health checks increase availability by replacing unhealthy instances. When you use multiple health checks, all are evaluated, and if at least one fails, instance replacement occurs.

EC2 health checks

Always enabled

Additional health check types - *optional* | Info

☑ Turn on Elastic Load Balancing health checks **Recommended**

Elastic Load Balancing monitors whether instances are available to handle requests. When it reports an unhealthy instance, EC2 Auto Scaling can replace it on its next periodic check.

EC2 Auto Scaling will start to detect and act on health checks performed by Elastic Load Balancing. To avoid unexpected terminations, first verify the settings of these health checks in the Load Balancer console

☐ Turn on VPC Lattice health checks

VPC Lattice can monitor whether instances are available to handle requests. If it considers a target as failed a health check, EC2 Auto Scaling replaces it after its next periodic check.

☐ Turn on Amazon EBS health checks

EBS monitors whether an instance's root volume or attached volume stalls. When it reports an unhealthy volume, EC2 Auto Scaling can replace the instance on its next periodic health check.

Health check grace period | Info

This time period delays the first health check until your instances finish initializing. It doesn't prevent an instance from terminating when placed into a non-running state.

300 seconds

EC2 > Auto Scaling groups > Create Auto Scaling group

Step 2
Choose instance launch options

Step 3 - *optional*
Integrate with other services

Step 4 - *optional*
Configure group size and scaling

Step 5 - *optional*
Add notifications

Step 6 - *optional*
Add tags

Step 7
Review

Group size Info

Set the initial size of the Auto Scaling group. After creating the group, you can change its size to meet demand, either manually or by using automatic scaling.

Desired capacity type

Choose the unit of measurement for the desired capacity value. vCPUs and Memory(GiB) are only supported for mixed instances groups configured with a set of instance attributes.

Units (number of instances)

Desired capacity

Specify your group size.

2

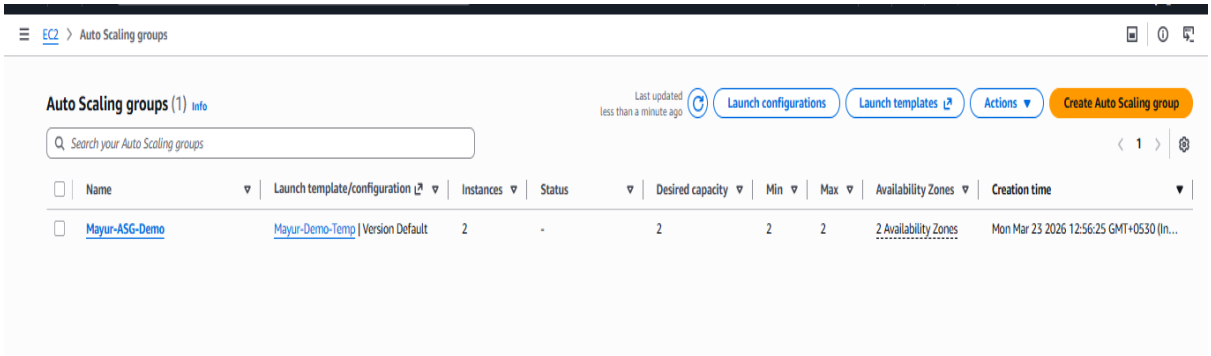
Scaling Info

You can resize your Auto Scaling group manually or automatically to meet changes in demand.

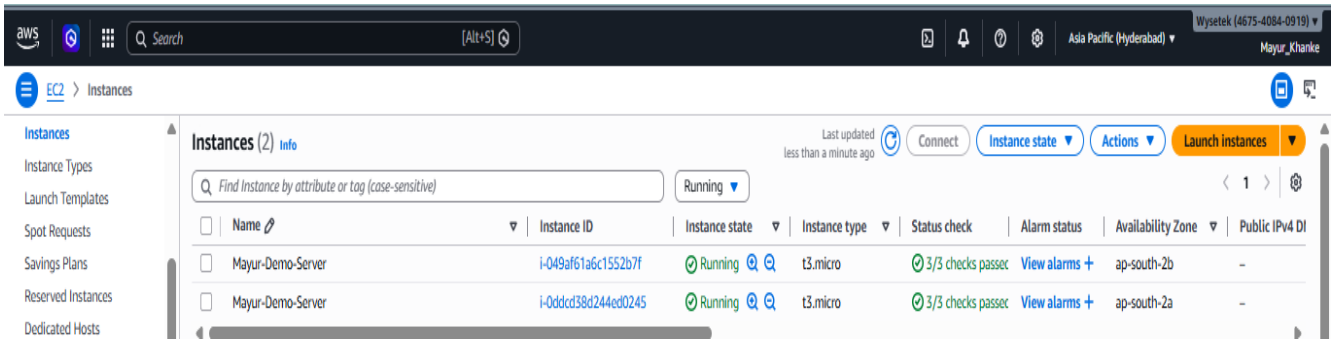
Scaling limits

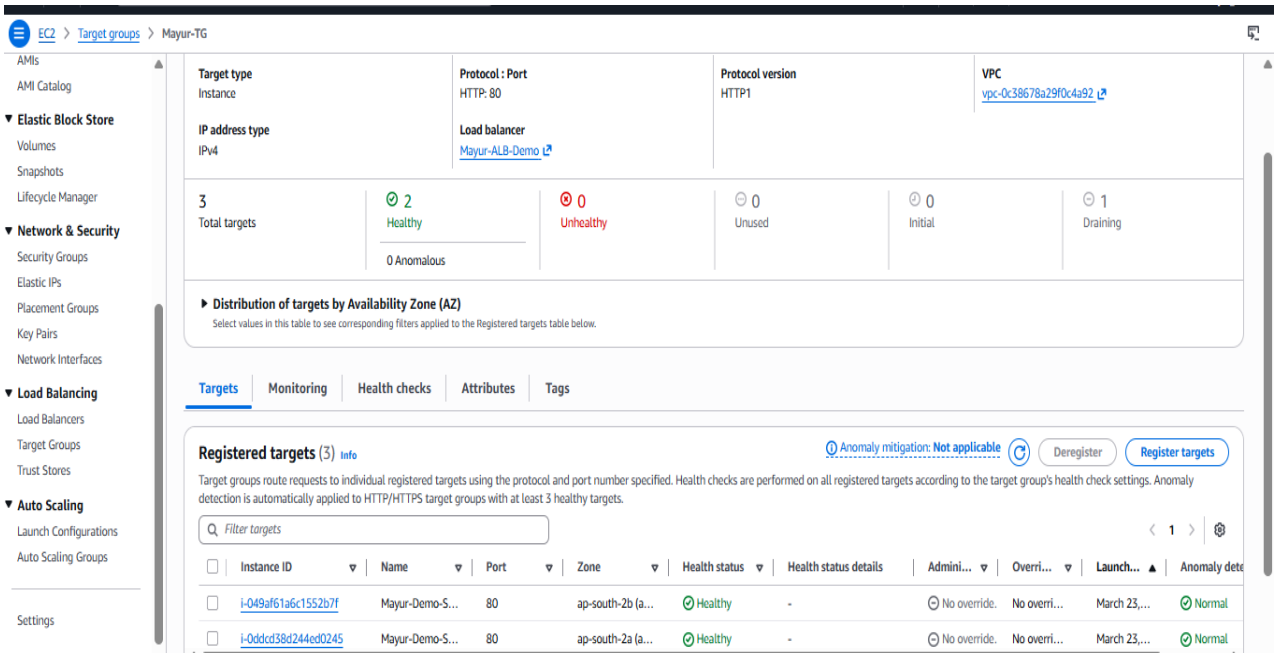
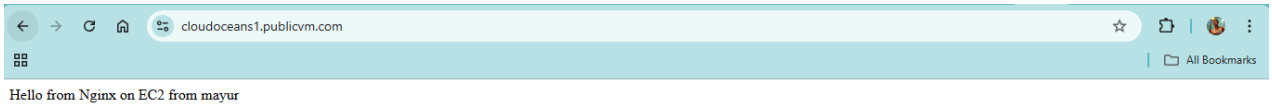
Set limits on how much your desired capacity can be increased or decreased.

Min desired capacity	Max desired capacity
2	2
Equal or less than desired capacity	Equal or greater than desired capacity



Step 11: - Instances will scale out to reach the desired capacity of 2.





Step 12: - When we access the domain name, the website will be displayed securely over an HTTPS request.